

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

CE 208

Quantity Surveying

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Cost Estimation for Different Materials in Structures



Lecture Outline

- Computation of Material Quantities
- Unit Weights of Different Structural Materials
- Schedule of Rates
- Cost Estimation

Computation of Material Quantities

- Step 1: Determine numbers and dimensions of members
- Step 2: Calculate their volumes
- Step 3: Apply correction factors for assumptions, shrinkage, and wastage
- Step 4: Differentiate materials
- Step 5: Convert the corrected volumes in appropriate units

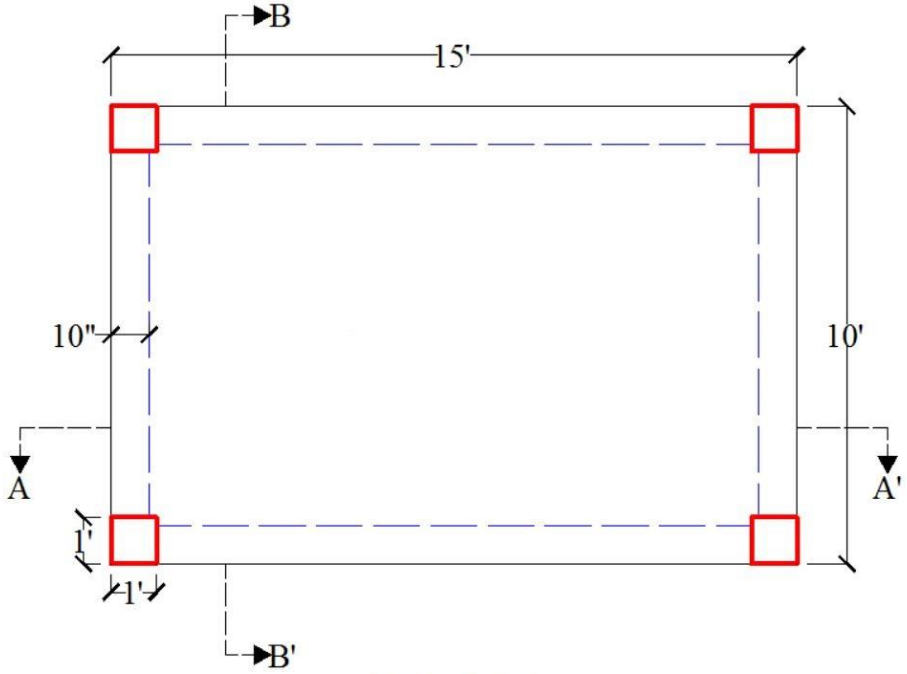
Computation of Material Quantities

- An one storied RCC building consists of 6” thick 10’X15’ rectangular floor slabs supported by beams at all of its edges. The beams are 10” wide and 12” deep supported by 4 columns of cross-section 12”X12” at the corners. Each column is supported by 4’X4’X1’6” footings at a depth of 5’ below plinth level. There are partition walls throughout the perimeter of the ground floor consisting of a door (3’X7’) and two windows (5’X4’). The parapet height is 2’6” and floor height is 10’. (Assume, reinforcement area = 1% of RCC section and concrete mix ratio = 1:2:4)

  Determine,

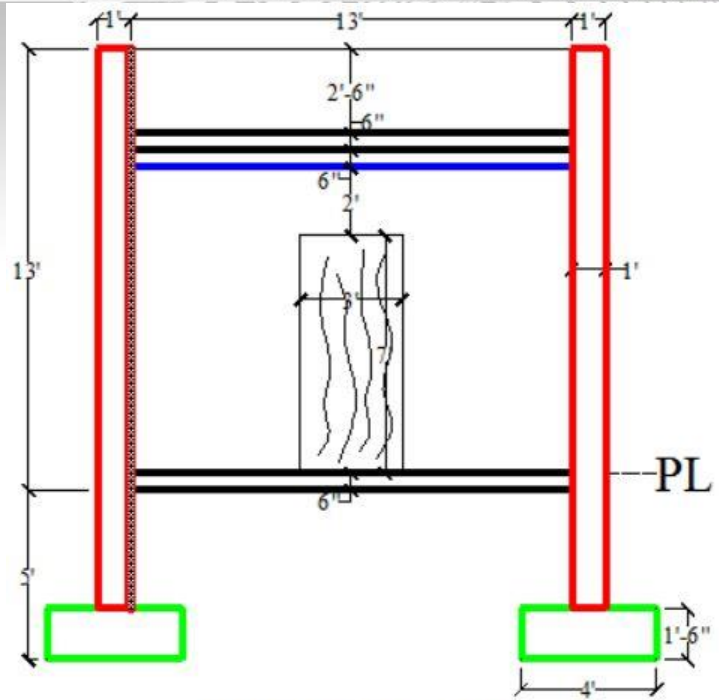
  - ✓ **The number of bags of cement required for construction.**
  - ✓ **The amount of coarse and fine aggregates required for construction.**
  - ✓ **The volume of steel required for construction.**
  - ✓ **The number of bricks required for construction.**

Computation of Material Quantities

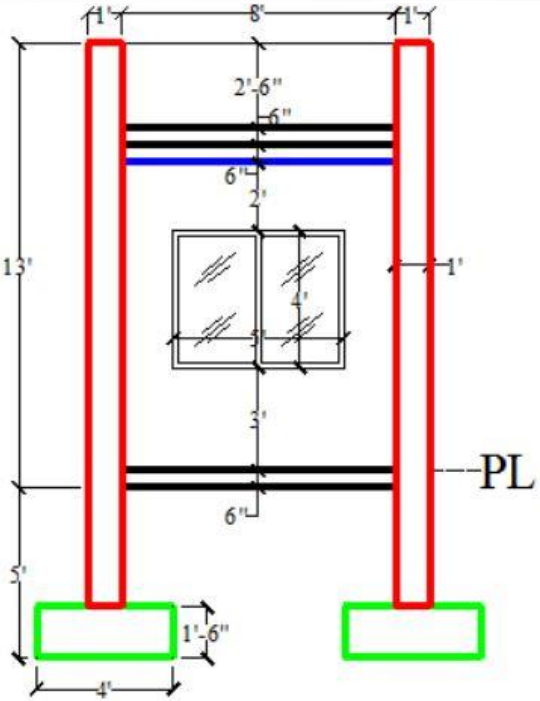


PLAN

Computation of Material Quantities



SECTION AA'



SECTION BB'

Unit Weights of Different Structural Materials

- Concrete – 150 pcf
- Steel – 490 pcf
- Brick – 120 pcf
- Water – 62.4 pcf
- Soil – 110~140 pcf

✓ **Determine the weights of all the materials in the previous problem.**

Schedule of Rates

- The costs associated with construction can be separated into –
  - ✓ Material costs – Market rates
  - ✓ Labour costs – PWD schedule of rates/Local contractor rates
- There will be additional **site specific costs** and **non-engineering costs** for specific projects

Cost Estimation

- Cement – 450~550 tk/bag
- Sand – 20~80 tk/cft
- Brick Chips – 80~100 tk/cft
- Stone Chips – 120~180 tk/cft
- Steel – 55000~80000 tk/ton
- Brick – 6000~9000 tk/thousand

✓ **The rates will vary according to the market.**

✓ **Determine the total cost of materials for the construction of the building in the previous problem.**



Thank You!